

simplicial subcomplex S_α^n homeomorphic to S^n at one end of $M(f_\alpha)$, and the homeomorphism $S^n \approx S_\alpha^n$ is homotopic in W_n to the map f_α , hence also to φ_α . Let Z_{n+1} be obtained from Z_n by attaching $D_\alpha^{n+1} \times I$'s via these homotopies between the φ_α 's and the inclusions $S_\alpha^n \hookrightarrow W_n$. Thus Z_{n+1} contains X^{n+1} at one end, and at the other end we have a simplicial complex $Y_{n+1} = Y_n \cup_\alpha C(f_\alpha)$, where $C(f_\alpha)$ is obtained from $M(f_\alpha)$ by attaching a cone on the subcomplex S_α^n . Since $D^{n+1} \times I$ deformation retracts onto $\partial D^{n+1} \times I \cup D^{n+1} \times \{1\}$, we see that Z_{n+1} deformation retracts onto $Z_n \cup Y_{n+1}$, which in turn deformation retracts onto $Y_n \cup Y_{n+1} = Y_{n+1}$ by induction. Likewise, Z_{n+1} deformation retracts onto $X^{n+1} \cup W_n$ which deformation retracts onto $X^{n+1} \cup Z_n$ and hence onto $X^{n+1} \cup X^n = X^{n+1}$ by induction.

Let $Y = \bigcup_n Y_n$ and $Z = \bigcup_n Z_n$. The deformation retractions of Z_n onto X^n give deformation retractions of $X \cup Z_n$ onto X , and the infinite concatenation of the latter deformation retractions is a deformation retraction of Z onto X . Similarly, Z deformation retracts onto Y . $\square$

Exercises

1. What is the minimum number of edges in simplicial complex structures K and L on S^1 such that there is a simplicial map $K \rightarrow L$ of degree n ?
2. Use the Lefschetz fixed point theorem to show that a map $S^n \rightarrow S^n$ has a fixed point unless its degree is equal to the degree of the antipodal map $x \mapsto -x$.
3. Verify that the formula $f(z_1, \dots, z_{2k}) = (\bar{z}_2, -\bar{z}_1, \bar{z}_4, -\bar{z}_3, \dots, \bar{z}_{2k}, -\bar{z}_{2k-1})$ defines a map $f: \mathbb{C}^{2k} \rightarrow \mathbb{C}^{2k}$ inducing a quotient map $\mathbb{CP}^{2k-1} \rightarrow \mathbb{CP}^{2k-1}$ without fixed points.
4. If X is a finite simplicial complex and $f: X \rightarrow X$ is a simplicial homeomorphism, show that the Lefschetz number $\tau(f)$ equals the Euler characteristic of the set of fixed points of f . In particular, $\tau(f)$ is the number of fixed points if the fixed points are isolated. [Hint: Barycentrically subdivide X to make the fixed point set a subcomplex.]
5. Let M be a closed orientable surface embedded in $\mathbb{R}^3$ in such a way that reflection across a plane P defines a homeomorphism $r: M \rightarrow M$ fixing $M \cap P$, a collection of circles. Is it possible to homotope r to have no fixed points?
6. Do an even-genus analog of Example 2C.4 by replacing the central torus by a sphere letting f be a homeomorphism that restricts to the antipodal map on this sphere.
7. Verify that the Lefschetz fixed point theorem holds also when $\tau(f)$ is defined using homology with coefficients in a field F .
8. Let X be homotopy equivalent to a finite simplicial complex and let Y be homotopy equivalent to a finite or countably infinite simplicial complex. Using the simplicial approximation theorem, show that there are at most countably many homotopy classes of maps $X \rightarrow Y$.
9. Show that there are only countably many homotopy types of finite CW complexes.